

UPSALA, ON, Canada

WMO: 714350

Lat: 49.0361N

Lon: 90.4614W

Elev: 1603

StdP: 13.86

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -28.2	(c) -22.8	(d) -34.9	(e) 0.8	(f) -27.4	(g) -29.5	(h) 1.1	(i) -22.1	(j) 16.1	(k) 9.1	(l) 14.3	(m) 8.4	(n) 3.1	(o) 310	(p) 0.612

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 21.4	(c) 82.6	(d) 67.5	(e) 79.4	(f) 65.4	(g) 76.6	(h) 63.5	(i) 70.6	(j) 79.2	(k) 68.2	(l) 75.8	(m) 66.2	(n) 73.4	(o) 7.8	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 67.5	(b) 107.2	(c) 75.5	(d) 65.4	(e) 99.7	(f) 72.7	(g) 63.5	(h) 93.0	(i) 70.2	(j) 35.5	(k) 79.4	(l) 33.5	(m) 76.1	(n) 31.8	(o) 73.4	(p) 78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.9	13.0	11.6	-35.9	87.1	6.5	2.3	-40.6	88.8	-44.4	90.1	-48.0	91.4	-52.7	93.1
			-35.8	73.0	5.9	2.1	-40.0	74.5	-43.4	75.7	-46.7	76.9	-51.0	78.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.3	2.5	7.2	19.9	34.5	48.2	58.7	63.1	60.8	53.3	39.2	24.5	10.4
	DBStd	23.51	14.47	13.58	13.41	9.96	8.60	6.98	6.02	6.31	8.48	8.67	11.40	13.55
	HDD50	6637	1472	1198	936	470	138	12	0	2	61	354	765	1229
	HDD65	10944	1937	1618	1400	915	524	210	109	161	361	801	1214	1694
	CDD50	1281	0	0	2	5	83	272	405	337	159	18	0	0
	CDD65	113	0	0	0	0	4	20	49	31	9	0	0	0
	CDH74	1240	0	0	0	1	82	277	493	313	73	1	0	0
	CDH80	224	0	0	0	0	14	50	102	51	7	0	0	0
Wind	WSAvg	5.1	5.1	5.2	5.3	5.4	5.4	4.9	4.9	4.7	4.9	5.3	5.3	5.0
Precipitation	PrecAvg	31.8	1.5	1.2	1.5	2.4	3.3	4.1	3.9	3.2	3.9	2.8	2.2	1.8
	PrecMax	39.4	2.8	2.4	3.1	5.9	5.9	7.8	6.6	5.4	7.8	6.1	4.5	3.3
	PrecMin	24.1	0.5	0.4	0.6	0.7	1.5	2.0	2.5	1.1	1.3	1.1	0.7	0.8
	PrecStd	3.6	0.5	0.5	0.7	1.0	1.1	1.4	1.1	1.1	1.6	1.1	1.0	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.3	43.5	57.6	67.4	81.9	85.3	87.1	84.9	80.3	69.4	54.6	39.1
		MCWB	32.9	37.6	47.9	50.4	62.7	69.2	70.4	70.1	67.6	58.7	46.7	35.1
	2%	DB	32.0	36.0	48.1	61.6	75.6	80.1	82.8	80.7	75.6	63.5	48.4	34.2
		MCWB	31.0	32.7	40.4	47.0	58.5	64.6	68.0	67.5	64.4	53.4	43.3	32.4
	5%	DB	28.8	32.3	42.7	56.5	70.7	77.0	79.3	77.4	71.8	58.4	43.0	32.0
		MCWB	27.5	30.0	35.9	44.2	55.2	61.7	66.0	65.7	62.5	51.6	39.1	31.1
	10%	DB	24.1	28.1	37.7	51.9	65.6	73.7	76.1	74.3	68.0	53.3	39.1	29.3
		MCWB	23.0	25.9	32.6	41.2	52.6	60.2	64.0	64.0	60.7	48.0	36.5	28.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.6	38.0	51.3	54.6	66.1	72.0	73.1	72.8	70.0	60.8	49.9	35.8
		MCDB	35.1	41.4	53.5	62.2	75.7	82.7	82.7	80.4	77.5	65.6	52.7	38.2
	2%	WB	31.0	33.1	41.3	49.0	61.7	67.8	70.7	69.9	66.8	55.9	43.8	32.9
		MCDB	32.1	35.3	47.0	58.7	70.7	76.2	79.4	78.5	72.9	61.5	47.8	34.0
	5%	WB	27.5	30.0	36.5	45.1	57.5	64.8	68.2	67.4	63.8	52.1	39.8	31.0
		MCDB	28.7	32.7	41.6	55.2	66.7	72.3	75.9	74.8	69.3	57.5	42.5	32.1
	10%	WB	23.0	25.9	33.0	41.9	54.2	62.5	66.2	65.4	61.2	48.3	36.6	28.1
		MCDB	24.0	28.0	36.8	50.8	63.6	70.2	73.5	72.1	66.9	52.6	39.0	29.2
Mean Daily Temperature Range	5% DB	MDBR	19.1	21.9	23.0	21.8	23.4	22.3	21.4	21.7	19.8	15.5	13.4	16.1
		MCDBR	16.7	20.9	23.6	29.9	31.5	28.1	25.3	25.3	24.8	25.0	17.3	15.4
		MCWBR	15.4	17.2	16.2	17.9	16.5	14.2	13.0	14.1	14.9	16.5	12.6	14.1
	5% WB	MCDBR	15.9	18.7	20.9	27.0	25.4	22.7	21.6	22.3	20.9	22.2	15.7	15.1
		MCWBR	14.9	15.7	15.4	17.4	14.9	13.2	12.7	13.7	13.7	15.9	12.2	13.8
Clear-Sky Solar Irradiance	taub		0.248	0.256	0.286	0.327	0.342	0.365	0.377	0.360	0.346	0.301	0.275	0.251
	taud		2.368	2.347	2.377	2.422	2.451	2.403	2.397	2.462	2.486	2.589	2.543	2.416
	Ebn at Noon		264	289	297	293	291	284	278	279	272	271	255	248
	Edh at Noon		21	27	31	33	34	36	36	32	28	22	18	17
All-Sky Solar Radiation	RadAvg		384	761	1168	1528	1671	1770	1814	1547	1116	651	344	266
	RadStd		55	81	98	149	95	140	114	126	90	90	37	30

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page